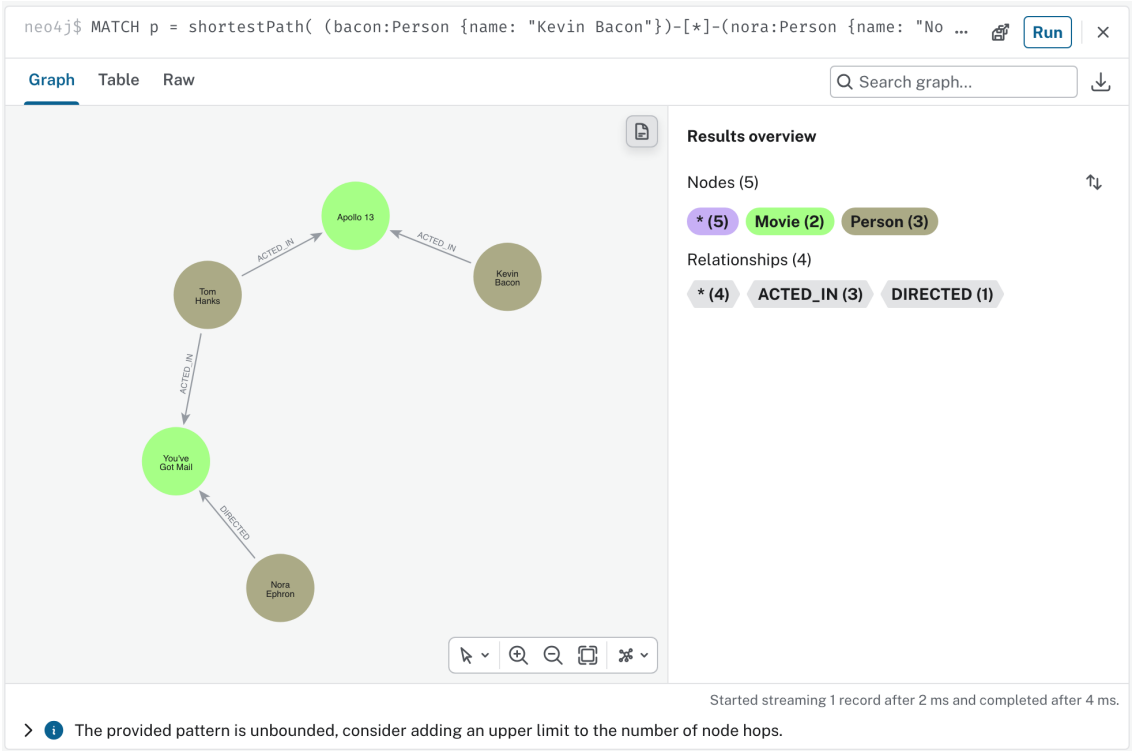
GraphDB (Neo4j): Academic Papers

Trevor Wattles · CS 452 · October 27, 2026

1. Bacon Path for Nora Ephron (Movies dataset)

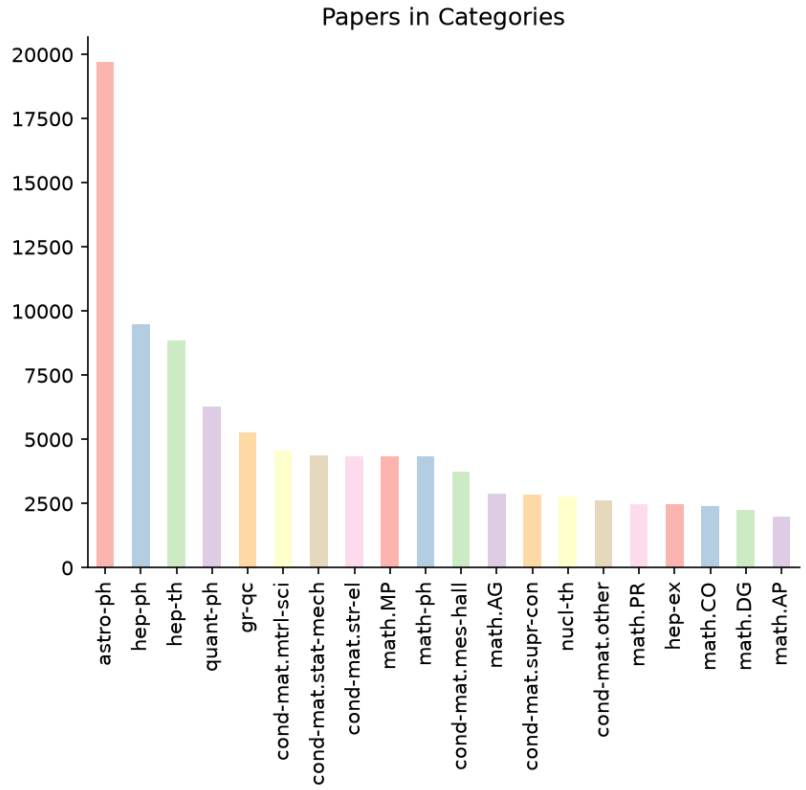
Screenshot from the Neo4j Browser (Neo4j 5 + APOC, running locally in Docker):




```
MATCH p = shortestPath(
  (bacon:Person {name: "Kevin Bacon"})-[*]-(nora:Person {name: "Nora Ephron"})
)
RETURN p
```

Path (length 4): (Kevin Bacon)-[:ACTED_IN]->(Apollo 13)<-[:ACTED_IN]-(Tom Hanks)-[:ACTED_IN]->(You've Got Mail)<-[:DIRECTED]-(Nora Ephron)

2. inDegree vs Category Name (top 20 categories)




```
MATCH (c:Category)
RETURN c.category, apoc.node.degree(c, "<IN_CATEGORY") AS inDegree
ORDER BY inDegree DESC LIMIT 20
```

3. Total number of papers in CS categories

5,812

unique CS papers (official answer)
count(DISTINCT p)

7,891

CS paper–category memberships
count(p): cross-listed papers counted once per CS category

For context: 100,000 papers total in the dataset, so 5.8% of all papers are in at least one CS category.

4. The queries

(A) Official answer: unique papers in at least one CS category

```
MATCH (p:Paper)-[:IN_CATEGORY]->(c:Category)
WHERE c.category STARTS WITH 'cs.'
RETURN count(DISTINCT p) AS uniqueCsPapers
```

Result: **5,812**

(B) Paper–category memberships (a paper in cs.AI and cs.LG counts twice)

```
MATCH (p:Paper)-[:IN_CATEGORY]->(c:Category)
WHERE c.category STARTS WITH 'cs.'
RETURN count(p) AS csCategoryMemberships
```

Result: **7,891**

(C) Cross-check with sum() over APOC in-degrees (equals B)

```
MATCH (c:Category)
WHERE c.category STARTS WITH 'cs.'
RETURN sum(apoc.node.degree(c, "<IN_CATEGORY")) AS sumCsInDegree
```

Result: **7,891**

5. Unique papers or once per CS category?

The **official answer is 5,812**, from query (A), which counts **unique** papers. `MATCH (p:Paper)-[:IN_CATEGORY]->(c:Category)` produces one row for *every* paper–category relationship, so a paper listed in several CS categories appears in several rows. `count(DISTINCT p)` removes those duplicates before counting.

Query (B) uses plain `count(p)`, so it counts a paper **once per CS category it is in** and gets 7,891. Query (C) sums the bar heights of the CS categories and gets the same number for the same reason. The difference of **2,079** comes from the 1,792 CS papers that are cross-listed in two or more CS categories (e.g. cs.LG + cs.AI).

6. Something fun: which CS categories are cross-listed together most?

```
MATCH (c1:Category)<-[:IN_CATEGORY]-(p:Paper)-[:IN_CATEGORY]->(c2:Category)
WHERE c1.category STARTS WITH 'cs.' AND c2.category STARTS WITH 'cs.'
  AND c1.category < c2.category
RETURN c1.category AS category1, c2.category AS category2, count(p) AS sharedPapers
ORDER BY sharedPapers DESC LIMIT 10
```

#	Category 1	Category 2	Shared papers
1	cs.DM	cs.DS	104
2	cs.CC	cs.DM	70
3	cs.CC	cs.DS	70
4	cs.IT	cs.NI	58
5	cs.DM	cs.IT	56
6	cs.AI	cs.NE	56
7	cs.CC	cs.LO	55
8	cs.AI	cs.LG	52
9	cs.DC	cs.NI	45
10	cs.CR	cs.IT	43

What this shows: the most common CS pairing is cs.DM (discrete math) + cs.DS (data structures & algorithms) with 104 shared papers, and the top three pairs are all theory (DM / DS / CC). cs.AI + cs.LG is only 8th (52 papers). That fits the data: this sample covers arXiv papers from April 2007 to December 2008, years before deep learning made cs.LG the biggest category on arXiv. Bonus: cs.IT is the largest CS category (1,487 papers) because every cs.IT paper is also tagged math.IT, the alias for the same Information Theory archive.

How it works: the pattern `(c1)<-[:IN_CATEGORY]-(p)-[:IN_CATEGORY]->(c2)` finds two category memberships of the same paper. `c1.category < c2.category` keeps each unordered pair once and removes self-pairs.